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ABSTRACT
Although solar-induced chlorophyll fluorescence (SIF) has been proven to be a useful indicator for dynamic drought monitoring, accurately quantifying the trajectory of vegetation SIF change during drought events remains a research challenge. In this case study, we applied a rapid change index (RCI) for the 8-day global 'Orbiting Carbon Observatory (OCO-2)' SIF product (GOSIF) to calculate the deviation between GOSIF trajectories and climate norms across China's Loess Plateau (LP) from 2001 to 2020. We compared the performances of SIF-RCI with GOSIF values and GOSIF anomalies during drought events and evaluated the drought warning potential with the widely used 8-day Standardized Precipitation Evapotranspiration Index (SPEI). Our results showed that SIF-RCI captured the rapid change areas in GOSIF caused by drought and the trajectory of rapid vegetation changes compared to GOSIF values and GOSIF anomalies. The SIF-RCI negative signal accurately identified the onset of drought almost simultaneously with SPEI. The SIF-RCI positive signal also predicted drought recovery with at least 4 8-day periods of lead time. Meteorological factors, especially precipitation, had a great impact on vegetation photosynthesis and this was detected with GOSIF. These impacts varied in different climate zones and vegetation types. Our study contributes to regional drought monitoring and assessment.

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Solar-chlorophyll fluorescence; drought warning; standardized precipitation evapotranspiration index; meteorological factors; vegetation

1. Introduction

Drought is becoming more extreme and frequent under the changing climate (Grillakis 2019; Zhao et al. 2020), posing a significant threat to food security and ecosystem stability globally (Mishra and Singh 2010). Drought is typically characterized by a severe lack of rainfall, runoff, or soil moisture (Palmer 1965; Mishra and Singh 2010), and the phenomenon of natural disasters with imbalanced

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water cycles in different regions. Drought is divided into four types, including meteorological, hydrological, agricultural, and socioeconomic. Unlike other extreme climate disasters, the process of drought formation is slow and complex, involving multiple aspects such as atmospheric changes, hydrological conditions, and ecosystems (Madadgar and Moradkhani 2014). Prolonged drought events can seriously affect the productivity and carbon cycle of terrestrial ecosystems (Li et al. 2022; Xiao et al. 2009). For example, from 2000 to 2009, global terrestrial net primary productivity (NPP) declined by 0.55 Pg because of prolonged and severe droughts, which reduced soil moisture availability and hindered plant photosynthesis, thereby affecting ecosystem productivity (Zhao and Running 2010). Increasing temperatures boost surface evapotranspiration, contributing to drought occurrence and intensification (Li and Xiao 2020). Simultaneously, reduced precipitation limits soil moisture replenishment, further increasing surface temperatures and evapotranspiration. Global warming exacerbates dryness, making droughts more intense and faster, bring great uncertainty in drought monitoring. Therefore, timely and accurate monitoring of drought occurrences is crucial for sustainable agricultural production and ecosystem stability.

The identification and monitoring of drought often relies on drought indicators calculated by remote sensing or meteorological data (Zhang et al. 2018). For different types of droughts in different regions, numerous drought indicators have been developed (Zhang et al. 2017). These drought indicators are quantitative indicators to describe the duration, intensity, and spatial drought distribution of regional level by aggregating one or more variable values into a single value (Quiring and Ganesh 2010; Ciais et al. 2005; Belayneh et al. 2014). The currently commonly used meteorological drought indicators comprise the Palmer Drought Severity Index (PDSI), the Standardized Precipitation Index (SPI) (McKee, Doesken, and Kleist 1993) and the standardized precipitation evapotranspiration index (SPEI). Compared to the other two indicators, SPEI is currently the most widely used meteorological indicator (Zhang et al. 2015, 2017). Despite the advantages, as the meteorological drought indexes are calculated based on the surrounding environment of vegetation, they may be insufficient to monitor terrestrial vegetation drought as drought can be affected by non-meteorological factors. Thus, considering the quality of vegetation growth and using meteorological indicators as references to evaluate drought may be more persuasive. Vegetation indices (VIs) based on remote sensing reflectance can provide successive observation of vegetation in time and space, and they are widely used in drought monitoring for large areas (Tian et al. 2019). However, these remotely sensed vegetation indices often only reflect the greenness of vegetation or changes in vegetation structure (Rossini et al. 2015). In contrast, vegetation photosynthesis is more sensitive and responsive to water stress conditions than structural response. Photosynthesis may decrease within a few days after water stress occurs, making it a more immediate indicator of drought, while structural changes such as leaf wilting or shedding typically take longer to manifest.

Vegetation photosynthesis, as indicated by Solar-induced chlorophyll fluorescence (SIF), has evolved as the primary monitoring tool for tracking vegetation growth conditions and water stress due to the advent of quantitative remote sensing technologies for vegetation observation in recent years (Sun et al. 2017; Shekhar et al. 2020; Li et al. 2022; Dang et al. 2021; Yao et al. 2022). After plants absorb light, excited chlorophyll molecules emit a light signal known as SIF that can accurately depict the dynamic changes that occur during actual photosynthesis in vegetation (Baker 2008; Liu, Dang, and Yue 2021). Diagnosing changes in SIF can extract information on changes in photosynthesis and productivity of regional vegetation ecosystems under water stress (Li and Xiao 2020). Drought affects the productivity and carbon cycle of the terrestrial ecosystem primarily by affecting photosynthesis, which is a major driver of the decline in global NPP (Chen et al. 2019). SIF responds and provides an early drought warning quicker than VIs when vegetation is subjected to drought stress (Tian et al. 2019; Liu et al. 2018; Yoshida et al. 2015), making it highly correlated with drought monitoring. Even if the VI (e.g. NDVI) remains unchanged (Daumard et al. 2010), SIF will decrease in drought, so satellite-derived SIF is the most sensitive signal describing dynamic changes in photosynthesis.

Due to its sensitivity to environmental stress, SIF has been frequently utilized to investigate how drought and heat wave events affect vegetation (Ma et al. 2018; Koren et al. 2018). For instance, SIF was applied in tracking the response of vegetation physiological processes on environmental pressure during widespread drought occurrences in many countries (Wang et al. 2016; Tian et al. 2019; Zhang et al. 2022), and the outcomes revealed that SIF performed better than other VIs (Hao et al. 2021; Helm et al. 2020; Li and Xiao 2019). Yang et al. (2018) utilized GOME-2 SIF data to evaluate the decrease in forest yield caused by drought in the Amazon Forest from 2015 to 2016. Moreover, SIF has been shown to have excellent ability to capture and quantify the gross primary productivity (GPP) losses brought on by drought (Li and Xiao 2019). These studies used SIF to monitor drought and almost all used SIF or SIF standardized anomalies to reflect the dynamic changes of vegetation. Even though they offered a fresh opportunity to monitor how drought affected ecosystem productivity, the water stress of vegetation in different regions under drought stress was different during drought and changed constantly with the development of drought. Highlighting rapid changes of vegetation under water stress and tracking abnormal changes in vegetation caused by drought is the focus of current research. Based on two flash drought events that occurred in the United States, Mohammadi, Jiang, and Wang (2022) explored the performance of SIFRCI in drought monitoring by quantifying the deviation between SIF trajectories and the climatological norm. They demonstrated that SIFRCI had the potential for early warning of flash drought. Their research provides important insights and data foundation for monitoring and early warning of flash drought. It is precisely because SIF-RCI has the advantage of capturing rapid changes in vegetation under water stress that we hope to explore its performance to evaluate whether it can perform better under general drought conditions. As one of the most seriously affected areas by soil erosion in the world, the ecological environment of the Loess Plateau (LP) was fragile and extremely vulnerable to climate change, and the vegetation was highly susceptible to water stress (Wang et al. 2015). Since the 'Grain for Green and grassland' project was put into action in 1999, a considerable portion of agricultural land had been transformed into forest and grassland, resulting in a significant increase in vegetation cover (Fu 2022). However, large-scale vegetation restoration also brought challenges to the water recycling system in the region. On the one hand, excessive water consumption by vegetation has led to extensive development of the dry soil layer (Liu et al. 2022). On the other hand, the LP's current vegetation productivity is approaching its threshold of water resource carrying capacity (Feng et al. 2016). Further expansion of the vegetation restoration area would inevitably exacerbate water shortages and other issues, posing a major threat to the long-term viability of ecosystems and agricultural productivity. As global warming leads to rising temperatures and increasing probability of drought occurrence (Breshears 2005), the drought situation on the LP becomes more complex and severe. Therefore, timely and accurate identification and drought monitoring on the LP, as well as analysis of how drought affects the vegetative ecosystem, are extremely crucial for future agricultural production and vegetation restoration.

In this study, based on the methods and findings of Mohammadi, Jiang, and Wang (2022), we aimed to further explore the applicability of SIFRCI in LP drought events. We will determine its reliability in LP drought monitoring through the comparison between the SIF-RCI with the widely used meteorological parameter SPEI from 2001 to 2020 across LP. The specific objectives of this research were: (1) to examine the drought warning potentials of SIF-RCI by comparing with the widely used meteorological indicator of SPEI; (2) to compare their performances in typical drought events with GOSIF (Orbiting Carbon Observatory (OCO-2) SIF product) value and GOSIF anomaly; and (3) to evaluate the impact of soil moisture and different meteorological factors on vegetation photosynthesis during drought.

2. Materials and methods

2.1. Study area

The LP is situated in the middle and upper portions of the Yellow River Basin (33°43′ – 41°16′ N, 100°54′ – 114 °33′ E), with 7 administrative provinces in the region from west to east. The total area of the LP is approximately 640,000 km^2, accounting for about 7% of China's land area, and it is the second largest plateau in China after the Qinghai-Tibet Plateau. The LP is characterized by a temperate continental monsoon climate with high temperatures and rainstorms in summer and a cold and dry winter (Fu 2022). From 2001 to 2020, the average annual precipitation and temperature of the LP is 454 mm and 8.6°C, respectively. According to the drought index proposed by the United Nations Environment Programme (UNEP), the LP was divided into four climate zones including arid (A), semi-arid (S-A), semi-humid (S-H) and humid (H) (Figure 1). The land use types of the LP include cropland, forest, grassland, construction land, water body and wasteland, which are shown in Figure 1 with the proportion of area occupied by various use types. Winter wheat and rice are the primary agricultural products. The landform of the LP is complex with the main landform types of loess hilly and gully region, loess tableland gully region, valley plain region, wind-blown sand region and soil and rock mountain region, and the soil and water loss are serious. The unique natural scenery with thousands of gullies, ravines, and fragmented terrain has gradually formed because of long-term erosion and exploitation of the ecosystem by human activities. The LP is mostly covered by thick layers of loess with loose soil and broken terrain, and it has been exposed to the hazards of soil erosion for a long time. In addition to soil degradation, soil erosion also triggered a series of serious ecological issues, including serious vegetation damage and poor ecological environment.

2.2. Datasets

2.2.1. Meteorological data

The meteorological data used in this study consisted of daily precipitation, temperature (maximum, average and minimum), air pressure, sunshine duration, wind speed and relative humidity provided by the National Meteorological Information Center (http://data.cma.cn/). These meteorological datasets were checked for quality and completeness. Considering the integrity of the data, some meteorological stations with invalid and missing data were discarded after screening and processing, and ultimately 201 meteorological stations within the LP were selected to create surface data covering the entire LP (Figure 1). To match the SIF data in time and space, these

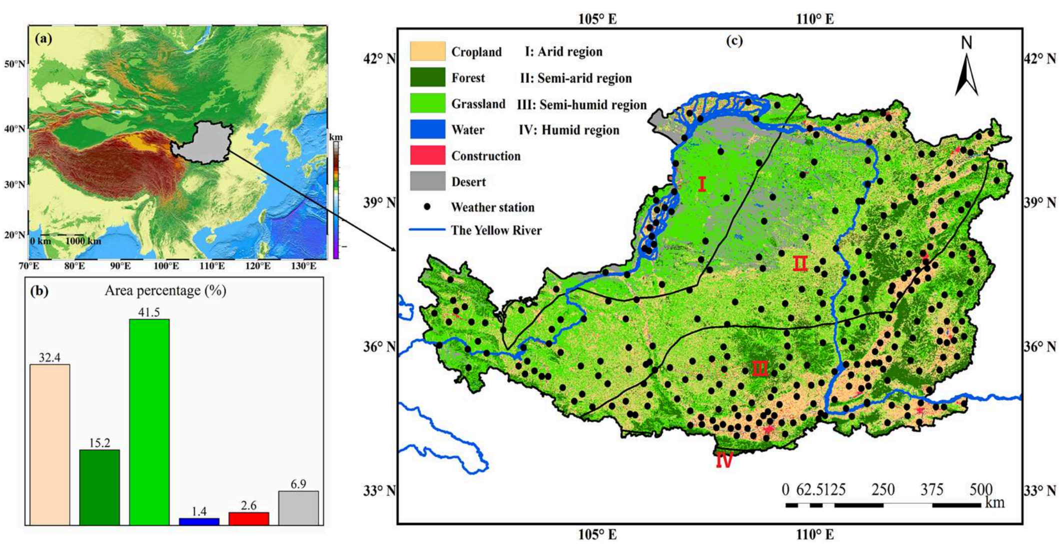

Figure 1. (a) The location of the study area, China's Loess Plateau; (b) The proportion of land use types; (c) Land use types and climate zoning.

meteorological datasets were spatially interpolated to a spatial resolution of 0.05° in ArcGIS 10.7 software at temporal resolutions of 8 days and one month.

2.2.2. MODIS data

The MODIS products used in this study include the MODIS-GPP and MODIS-ET derived from the data products of MODIS MOD17A2HGF and MOD16A2GF with 500 m spatial resolution. The MODIS-GPP product is applied to evaluate the loss of the gross primary productivity (GPP) caused by droughts, while the MODIS-ET product is used to evaluate the status of water loss. The temporal resolution of both MODIS-GPP and MODIS-ET data products is 8 days (data source: https://lpdaacsvc.cr.usgs.gov). The Photosynthetic Active Radiation (PAR) dataset used in this study was obtained from the Global Land Surface Satellite (GLASS) products (http://www.glass.umd.edu) with a spatial resolution of 0.05° at a daily step.

2.2.3. Satellite SIF

As an associated product of photosynthesis of light energy on leaves, SIF is a signal emitted by the chlorophyll in plants after absorbing solar energy and contains a large amount of photosynthetic information (Lichtenthaler et al. 1986; Genty, Briantais, and Baker 1998), which is considered as a fast and non-destructive indicator that can effectively estimate plant photosynthesis (Jonard et al. 2020). The disadvantages SIF data being directly measured from satellite sensors are discontinuity and coarse resolution. So, in this study data from the SIF product of GOSIF was used. This has high spatial and temporal resolution (i.e. 0.05° at 8-day, monthly and annual temporal resolutions) derived from Moderate Resolution Imaging Spectroradiometer (MODIS) and MERRA-2 meteorological reanalysis data (Li and Xiao 2019). GOSIF has been proven to play an important and unique role in research topics closely related to ecosystems such as water stress monitoring and vegetation productivity estimation (Song et al. 2018), which has been widely used in recent years (Castro et al. 2020; Xu et al. 2020). In this study, the 0.05° GOSIF data with 8-day and monthly time resolutions were used to assess the performances of ecosystem activities in drought monitoring (data source: https://globalecology.unh.edu).

2.2.4. Soil moisture data

The soil moisture (SM) data is from the global time-series surface soil moisture dataset with 1-km spatial resolution from 2001 to 2020 (Zheng, Jia, and Zhao 2022). This dataset first utilized surface soil water parameters of ERA5 (fifth generation ECMWF atmospheric reanalysis of the global climate) reanalysis data to produce ESA-CCI (European Space Agency Climate Change Initiative) surface soil water products with a spatial resolution of 0.25° at a daily time step. The spatial resolution of the daily surface soil moisture dataset was further downscaled to 1 km using the global soil moisture observation data from the international soil moisture observation network (ISMN) and a machine learning algorithm (Zheng, Jia, and Zhao 2023). Verification using 2346 ground observation sites around the world revealed that this product had a good accuracy with a correlation coefficient of 0.89.

To reduce the errors brought by inconsistencies in spatiotemporal resolutions, all these variables (including the GPP, ET, PAR and SM) were resampled to a 0.05° spatial resolution at an 8-day time step to guarantee conformity with the SIF product's temporal and spatial resolutions.

2.3. Analytical methods

Figure 2 depicts the technical flowchart, which includes data collection, preprocessing, and computational analysis. During data preprocessing, we conducted quality control on the data. For missing values caused by cloud occlusion or data collection issues, we used spatial interpolation and data fusion techniques to improve the quality of the data. The data processing mainly includes the calculation of drought indices (SPEI, GOSIF standardized anomaly and SIF-RCI),

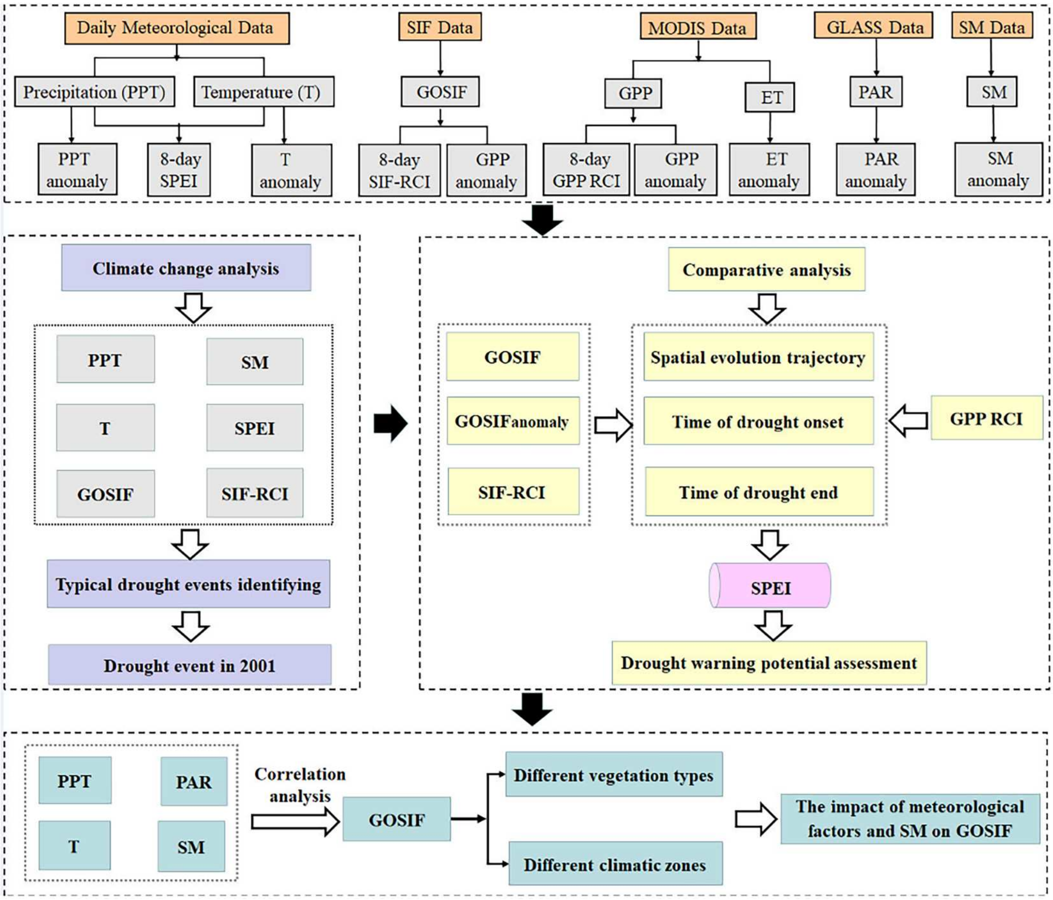

Figure 2. The technical flow chart for data collection, preprocessing and computational analysis was used in this study.

comparative analysis of GOSIF values, GOSIF anomalies and SIF-RCI, correlation analysis between hydrometeorological indicators and SIF.

2.3.1. Standardized anomaly

The standardized anomaly is used to reflect the drought signal, and the variables that need to be calculated for anomaly include SIF, GPP, precipitation, temperature, evapotranspiration, and soil moisture. The formula of standardized anomaly is as follows:

$$Y(p,q,t)\_anomaly = \frac{Y(p,q,t) - \overline{Y}(p,q)}{std(Y(p,q))} \tag{1}$$

where $Y(p,q,t)\_anomaly$ is the standardized anomaly at time t at the positions of each variable pixel (p,q); $Y(p,q,t)$ is each variable pixel's initial value at time t; $\overline{Y}(p,q)$ is the average value of each variable from 2001 to 2020 at pixel (p,q) positions; $std(Y(p,q))$ is the standard deviation of each variable from 2001 to 2020 at pixel (p,q) positions.

2.3.2. Standardized precipitation evapotranspiration index (SPEI)

The SPEI is calculated based on the moisture balance between precipitation and atmospheric evaporative demand (Vicente-Serrano, Beguería, and López-Moreno 2010). The SPEI for each 8-day period is calculated as follows:

First, the potential evapotranspiration (PET) (mm) for every 8 days is calculated by the Penman-Monteith (PM) model (Zhao et al. 2015):

$$PET_0 = \frac{0.408\Delta(R_n - G) + \gamma\frac{900}{T + 273}\mu_2(e_a - e_d)}{\Delta + \gamma(1 + 0.34\mu_2)} \tag{2}$$

where$PET_0$is the potential evapotranspiration (mm d^{-1});$R_n$is the net irradiance (MJ m^{-2} d^{-1}), and G is the soil heat flux density (MJ m^{-2} d^{-1}); T is the average air temperature (°C); $\Delta$is the slope of the saturation vapor pressure and temperature curve at a certain point (kPa°C^{-1});$\gamma$is the dry wet constant; $\mu_2$is the wind speed at the height of 2 meters from the ground (m s^{-1}); $e_a$is the saturated water vapor pressure (kPa), and$e_d$is the actual water vapor pressure (kPa).

Second, the difference between 8-day total precipitation (P_j) and 8-day total potential evapotranspiration (PET_j) is calculated as:

$$D_j = P_j - PET_j \tag{3}$$

where $D_j$is the difference between potential evapotranspiration and precipitation.

Third, a three-parameter log-logistic probability distribution function is used for calculating best fit on the calculated D_j sequence:

$$f(x) = \frac{\beta}{\alpha}\left(\frac{x-\gamma}{\alpha}\right)^{\beta-1}\left[1 + \left(\frac{x-\gamma}{\alpha}\right)^{\beta}\right]^{-2} \tag{4}$$

where α, β andγ are scale parameter, shape parameter, and position parameter, respectively and they are calculated by the following formulas:

$$\alpha = \frac{(\omega_0 - 2\omega_1)}{\Gamma(1 + 1/\beta)\Gamma(1 - 1/\beta)} \tag{5}$$

$$\beta = \frac{2\omega_1 - \omega_0}{6\omega_1 - \omega_0 - 6\omega_2} \tag{6}$$

$$\gamma = \omega_0 - \alpha\Gamma(1 + 1/\beta)\Gamma(1 - 1/\beta) \tag{7}$$

$\Gamma(\beta)$is a gamma function of β; ω_0, ω_1, and ω_2 are probability weights; $F(x)$ the cumulative probability distribution function can be calculated by the following equation:

$$F(x) = \int_0^x f(t)dt = \left[1 + \left(\frac{\alpha}{x - y}\right)^{\beta}\right]^{-1} \tag{8}$$

Finally, the cumulative probability is converted into SPEI:

$$SPEI = \omega - \frac{C_0 + C_1\omega + C_2\omega^2}{1 + \omega d_1 + \omega^2 d_2 + \omega^3 d_3} \tag{9}$$

$$\omega = \sqrt{-2\ln P}, P \leq 0.5 \tag{10}$$

where, $P = F(x)$, when $P > 0.5$, using (1-P) instead of P; $C_0 =$ 2.515517, $C_1 =$ 0.802853, $C_2 =$ 0.010328, $d_1 =$ 1.432788, $d_2 =$ 0.189269, $d_3 =$ 0.001308.

The SPEI is a standardized drought indicator with multiple timescales, and the SPEI of different timescales indicate the cumulative water deficit, while SPEI at short time scales is more sensitive to soil moisture response (Ling et al. 2023). Therefore, to maintain synchronization with GOSIF data over time, we used an 8-day scale SPEI in this study. Table 1 is the drought level classification, which is categorized in accordance with the national meteorological drought level standard released in 2017.

Table 1. SPEI drought level classification.

Level	Type	SPEI
1	Normal	(>−0.5)
2	Slight drought	(−1.0.−0.5)
3	Moderate drought	(−1.5.−1.0)
4	Severe drought	(−2.0.−1.5)
5	Extreme drought	(≤−2.0)

2.3.3. Rapid change index (RCI)

Because anomalous weather patterns may last for several weeks and lead to the development or recovery of drought, the abnormal change rate of SIF over an extended period can indicate the state of water stress and vegetation health. To identify areas undergoing rapid changes in GOSIF, we calculated the RCI of 8-day GOSIF data based on the RCI formula of Otkin et al. (2014). Unlike Mohammadi, Jiang, and Wang (2022), who used the 8-d SIF data from GOME-2 to calculate the RCI, we selected GOSIF data for the last 20 years to calculate the RCI in this study, which was represented by SIF-RCI.

The standardized anomalies of the temporal change rate of GOSIF needs to be calculated before computing RCI and the formula is as follows:

$$\Delta V(K_1, K_2, A) = \frac{[V(K_2, A) - V(K_1, A)] - \frac{1}{N}\sum_{y=1}^{N}[V(K_2, A) - V(K_1, A)]}{\sigma(K_1, K_2)} \quad (11)$$

where K_1 and K_2 are the two 8-days used to calculate the difference and $V(K, A)$ is the GOSIF composite value for the Kth 8-day in year A. The first and second terms in the numerator represent the change rate between 8-days of K_1 and K_2 over N years and the climatological average rate of change within N years, respectively. The denominator is the SD of the change rate over N years. A negative ΔV denotes that the GOSIF increases slowly or decreases faster than usual, while a positive ΔV means the opposite.

When the ΔV value exceeds or falls below a specific threshold the RCI only increases or decreases within the subsequent 8-day period. To identify regions with exceptionally significant changes in GOSIF, the threshold was set at 0.75. The RCI represents the accumulation of ΔV beyond the set threshold of 0.75, the computed formula is as follows:

$$RCI = RCI_{prev} - \sqrt{abs(\Delta V) - 0.75} \; if \; \Delta V < -0.75 \quad (12)$$

$$RCI = RCI_{prev} + \sqrt{abs(\Delta V) - 0.75} \; if \; \Delta V > 0.75 \quad (13)$$

Where RCI_{prev} is the previous 8-day's RCI value. The RCI value resets to zero when the sign of ΔV differs from that of the previous 8-day period. However, it remains unchanged if ΔV keeps the same sign within the previous 8-day period but its magnitude falls below the specified threshold of 0.75. During the growing season, the negative RCI values correspond to an imminent onset of drought, while positive values indicate an approaching recovery from drought (Mohammadi, Jiang, and Wang 2022).

To compare the dynamic performance of satellite-derived variables related to vegetation function in capturing drought, we also applied the calculation formula of RCI to the 8-day GPP from MODIS.

2.3.4. Accuracy assessment of drought early warning

To compare the drought warning potentials of GOSIF value, GOSIF anomaly and SIF-RCI in LP, we conducted a correlation analysis between SPEI and GOSIF values, GOSIF anomaly and SIF-RCI, respectively. To evaluate the accuracy of SIF-RCI as a warning for the LP, we used the SIF-RCI value with the strongest signal, whether it was positive or negative, from May to June each year from 2001 to 2020 in the LP to correlate it with the maximum GOSIF value representing ecosystem productivity indicators and the strongest SPEI value (positive or negative) of the meteorological drought indicator during the peak growth season. We selected two sample areas with the size of $1^\circ \times 1^\circ$ (39-40° N, 111-120° E; 36-37° N, 110-111° E) that were severely affected by drought for analysis. To minimize the impact of non-uniformity in all grid data on the final results, we obtained the Pearson correlation coefficient and least squares linear regression line by using the spatial averages of the data across each sample area from 2001 to 2020, including the strongest SIF-RCI signal, the maximum GOSIF, and the strongest SPEI signal from May to June. The highest SPEI value between July and August corresponds to the strongest positive SIF-RCI signal between May and June, and vice versa. Thus, the robustness of SIF-RCI as a warning indicator can be tested by assessing the predictive relationship between the strongest SIF-RCI signal and two benchmark indicators of the maximum GOSIF and SPEI values.

2.3.5. Correlation analysis

The Pearson correlation analysis method was also applied to analyze the correlation between GOSIF and soil moisture (SM) and meteorological factors of precipitation (PPT), PAR, and temperature (T) to explore the regulatory role of meteorological factors on vegetation activities. The following is the formula for calculating the Pearson correlation coefficient r:

$$r = \frac{n\sum f_i g_i - \sum f_i \sum g_i}{\sqrt{\left[n\sum f_i^2 - (\sum f_i)^2\right]\left[n\sum g_i^2 - (\sum g_i)^2\right]}} \tag{14}$$

where $-1 \leq \mathrm{r} \leq 1$and when $r > 0$, it indicates the relationship between f and g is positively correlated, otherwise it is negatively correlated. When $|r|$approaches 1, there is a significant correlation between f and g but when $|r|$ approaches 0 there is a weak correlation between f and g.

3. Results

3.1. Climate change conditions over the past 20 years

Figure 3(a) shows the monthly average climate change on the LP from 2001 to 2020. According to the changes in monthly precipitation over the last 20 years, precipitation across the LP remained consistently low. Monthly precipitation was generally below 180 mm, especially before 2011. Precipitation during summer was less than 140 mm. The highest monthly rainfall in most years was about 100 mm. SM data were responsive to precipitation and follow similar patterns as precipitation changes. The soil moisture in 2001, 2007, 2011 and 2013 was the lowest during the 20-year period, especially in spring from March to May. The soil moisture in summer increased with the rise in rainfall. From the temperature changes in Figure 3(a) and the trend analysis of temperature data from 2001 to 2020 (Figure S1), the temperature of the LP has not changed significantly over the previous 20 years. Figure 3(b) shows the changes of monthly mean SPEI from 2001 to 2020 in the LP, indicating wet and dry conditions. From the SPEI trends as shown on Figure 3(b), several drought events have occurred on the LP in 2001, 2006, 2013, 2015, and 2017. The portion of areas with negative SIF-RCI signals to the entire LP was calculated for each 8-day interval. The results showed that the total percentage of negative SIF-RCI every 8 days in 2001 was 8.0, the highest in the last 20 years (Figure 3(c)). Therefore, we selected the drought events that occurred on the LP in 2001 to assess the early warning impact of SIF-RCI on drought as an example.

3.2. Spatiotemporal evolution of drought in 2001 across the LP

Figure 4 displays the spatiotemporal change trajectory of the 8-day SPEI on the LP from April to September of 2001. In 2001, abnormal dryness spread throughout various regions of the LP. In

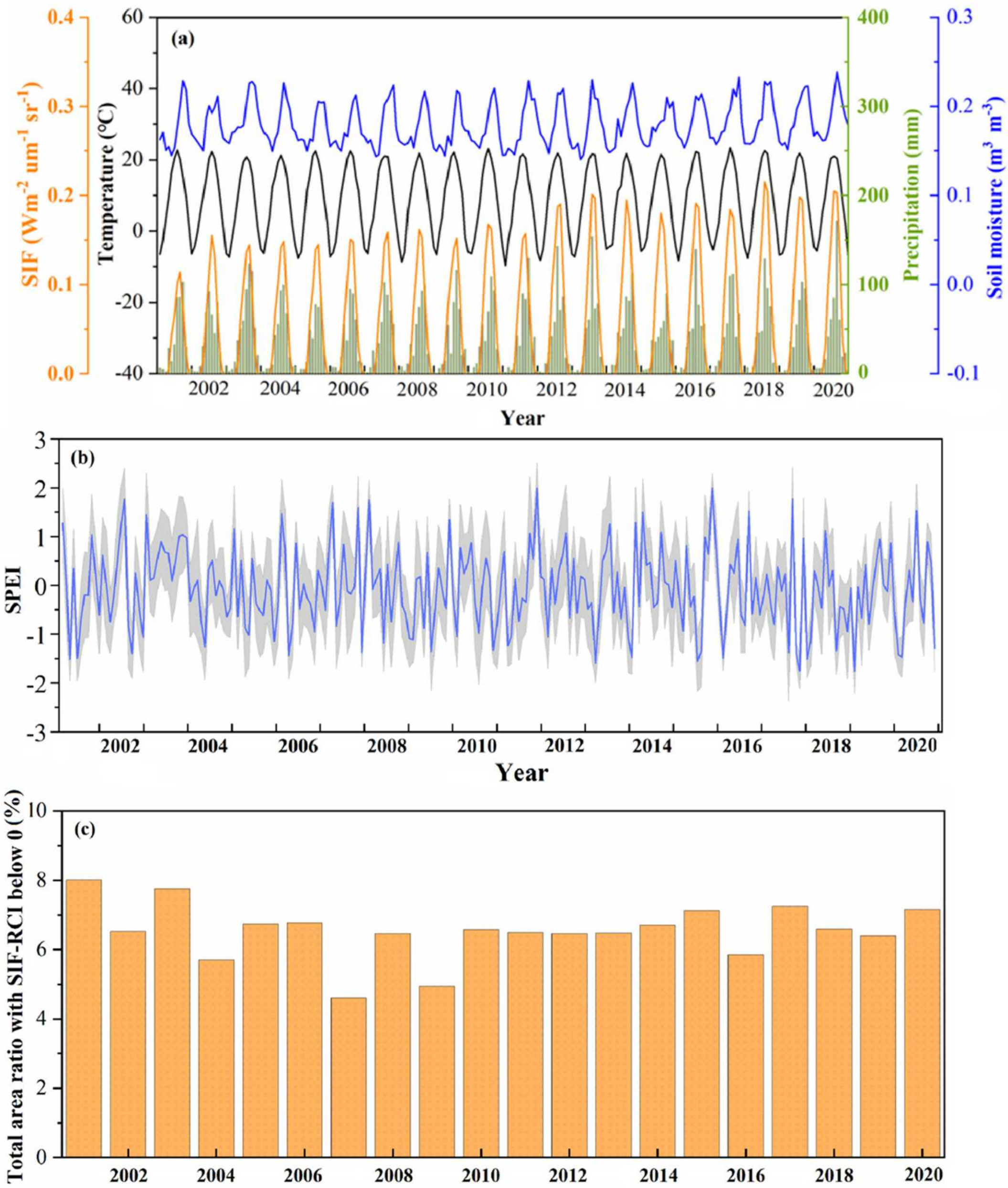

Figure 3. Climate changes from 2001 to 2020 over the LP: (a) the monthly average changes in solar-induced chlorophyll fluorescence (SIF), air temperature (T), precipitation (PPT) and soil moisture (SM); (b) the SPEI changes with ranges within ±1 standard deviation indicated in light gray shades; and (c) the total percentage of negative SIF-RCI every 8 days.

April, most areas of the LP began to experience abnormal drought, which reached a severe level (Level 4 in Table 1) in early May, and the signal of extreme drought began to appear. From 09 May to 03 June, the S-A and S-H zones experienced a rapid worsening of the drought, which peaked on 25 May. Subsequently, the drought intensity began to significantly weaken and the spatial extent decreased. Drought was mainly distributed in the zones of H and S-A, which almost completely disappeared in mid-August. Figure 5 depicts the monthly changes of SPEI, PPT, T, ET, SM, and GOSIF from March to September and the drought level of 2001 among the past 20 years. Except

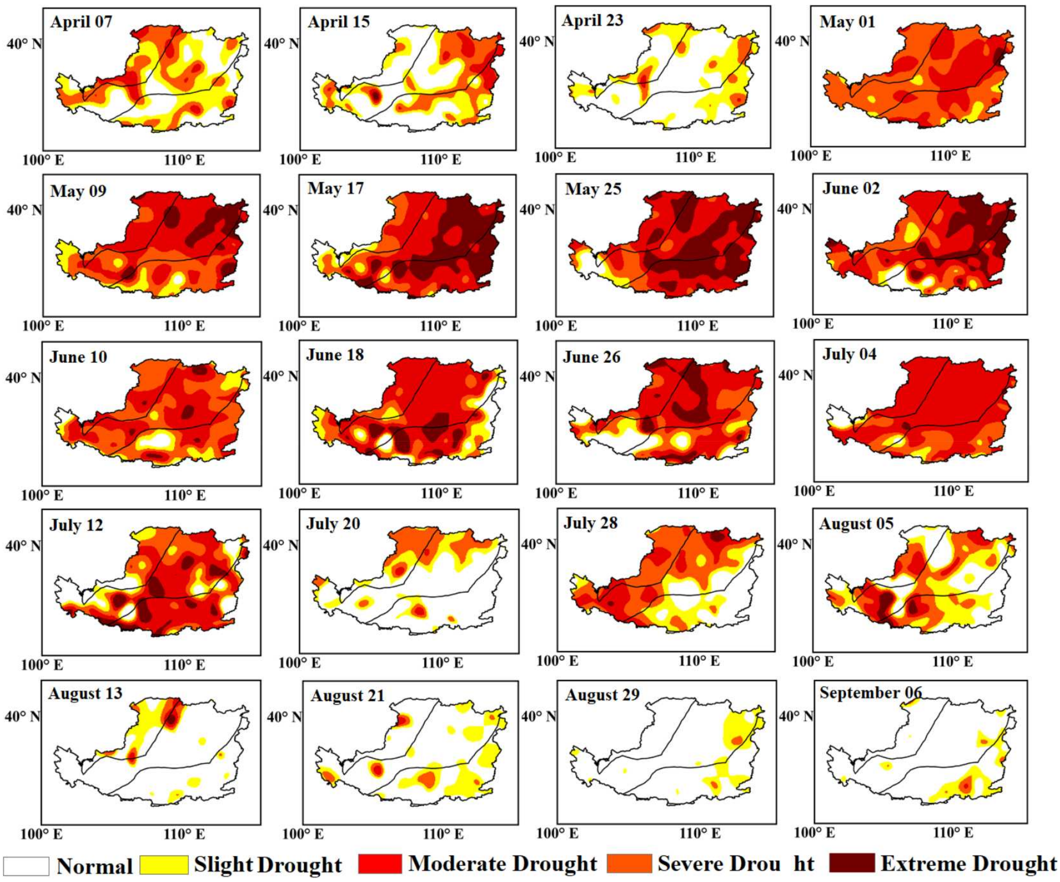

Figure 4. Spatial distributions of 8-day SPEI drought categories during the period 07 April to 06 September in 2001.

for SPEI and T, the PPT, SM, GOSIF, and ET from April to August 2001 were all lower than average from 2001 to 2020. The trend of changes in SPEI, SM, and PPT was largely consistent, and both had the lowest values in May which corresponded to the peak drought intensity in May.

3.3. Spatial distribution of GOSIF value, GOSIF anomaly and SIF-RCI across the LP

GOSIF values increased gradually with time during the growing season, especially in H and S-H zones of LP (Figure S2). However, vegetation growth in the H and S-H zones was better than that in the A and S-A zones regardless of drought. Therefore, vegetation growth changes caused by drought were difficult to identify. From the spatial distribution of GOSIF anomaly, vegetation growth was almost always below the normal level during the entire growing season (Figure S3). Drought first began in A and S-A zones in the early stages of the growing season. In May, the drought level in the boundary region between A and S-A zones was severe, weakening after peaking on 25 May. By 06 September, the drought had eased significantly with parts of the LP already above normal levels. In comparison with GOSIF value the GOSIF anomaly assessed the growth quality of vegetation by the deviation from normal level but the dynamic information on the time change rate of vegetation photosynthesis during drought was not clearly recognized.

Unlike GOSIF values and GOSIF anomalies, the SIF-RC highlighted the rapidly changing areas in GOSIF that were affected by drought (Figure 6). Except for the A zone, the negative SIF-RCI appeared in the majority of the LP's regions in early April. Although the negative value was

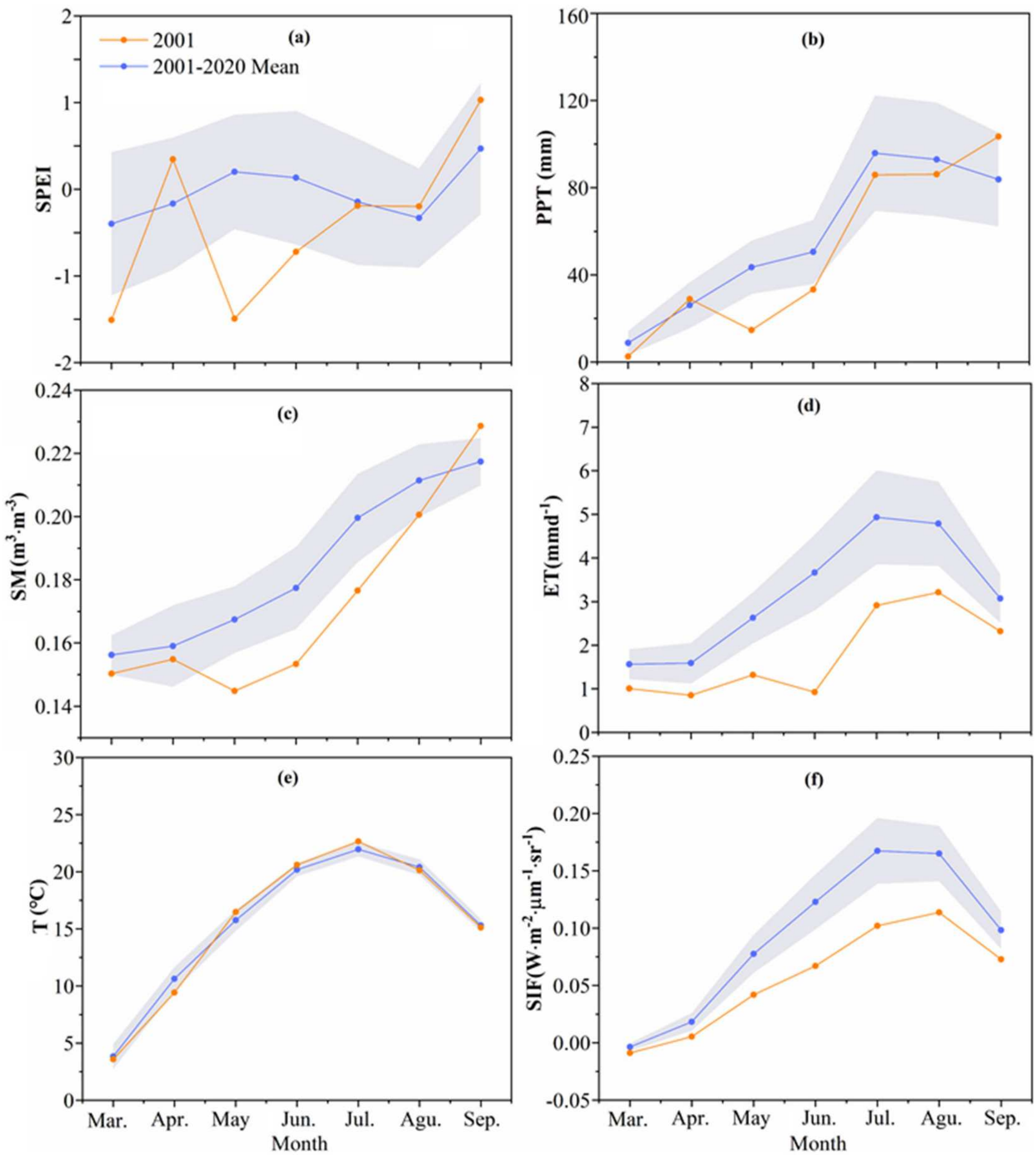

Figure 5. Seasonal variations in SPEI(a), PPT(b), SM(c), ET(d), T(e) and GOSIF(f) from March to September of 2001 in the LP. The gray shaded areas represent ±1 standard deviation of the multi-year monthly average.

small, it covered a wide area, particularly in regions dominated by croplands and forests, which are more sensitive to drought stress. From the end of April, the SIF-RCI values in the semi-humid (S-H) and humid (H) zones, especially in forested areas, decreased rapidly, showing significant negative values and a continuous expansion in area, particularly in the H zone where forests dominate. But the negative SIF-RCI signals in the H zone had completely disappeared in mid-May. The strongest negative signals of SIF-RCI occurred on 25 May in the S-A and S-H zones, and they decreased gradually with the persistence of drought and almost completely disappeared in early August. Thus, the SIF-RCI signal changed synchronously with the drought intensity and influence range monitored by SPEI.

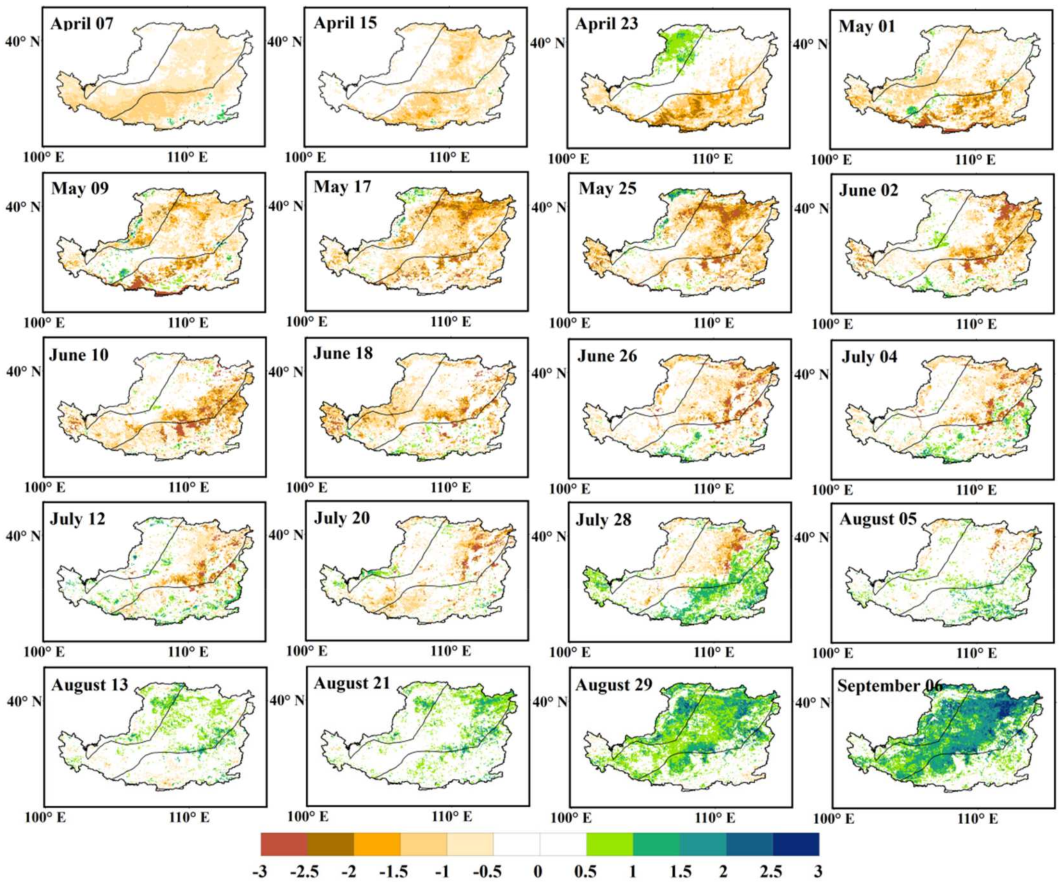

Figure 6. Temporal evolution of 8-day SIF-RCI values during 07 April to 06 September in 2001.

The positive RCI demonstrate an unusually rapid increase in photosynthetic activity indicating that vegetation growth is gradually recovering from the drought. The positive SIF-RCI signals were significantly enhanced, and the coverage area gradually increased after the drought intensity weakened significantly from the end of July. This demonstrates that SIF-RCI and SPEI drought signals are consistent and synchronous in drought warning on the LP.

Figure 7 showed that GPP RCI was inconsistent with the trajectory variation of drought signals detected by SPEI, nor did it present the pre-drought strong signal as detected by SIF-RCI in April 2001. During the drought indicated by SPEI, GPP-RCI signals were in general weaker and more spatially dispersed than SIF-RCI signals. Although GPP RCI showed positive signals in general, the spatial patterns were inconsistent with SPEI and did not show the trends as well as that of SIF-RCI. Therefore, SIF-RCI is more reliable than GPP RCI in indicating drought warning.

3.4. Comparison between SPEI and GOSIF value, GOSIF anomaly, and SIF-RCI

The correlation coefficients between SPEI and GOSIF value, GOSIF anomaly, and SIF-RCI are –0.26, 0.68, and 0.8, respectively (Figure 8), with the highest correlation between SPEI and SIF-RCI. Furthermore, compared with GOSIF value and GOSIF anomaly, SIF-RCI tracked the development of drought by highlighting areas undergoing rapid changes in GOSIF, demonstrating great potential as a predictor of the occurrence and recovery of drought.

The correlation coefficient between SPEI and GPP RCI in the LP was 0.5, which was significantly lower than between SPEI and SIF-RCI ($r = 0.80$) (Figure 8(a)). Although the GPP RCI value also

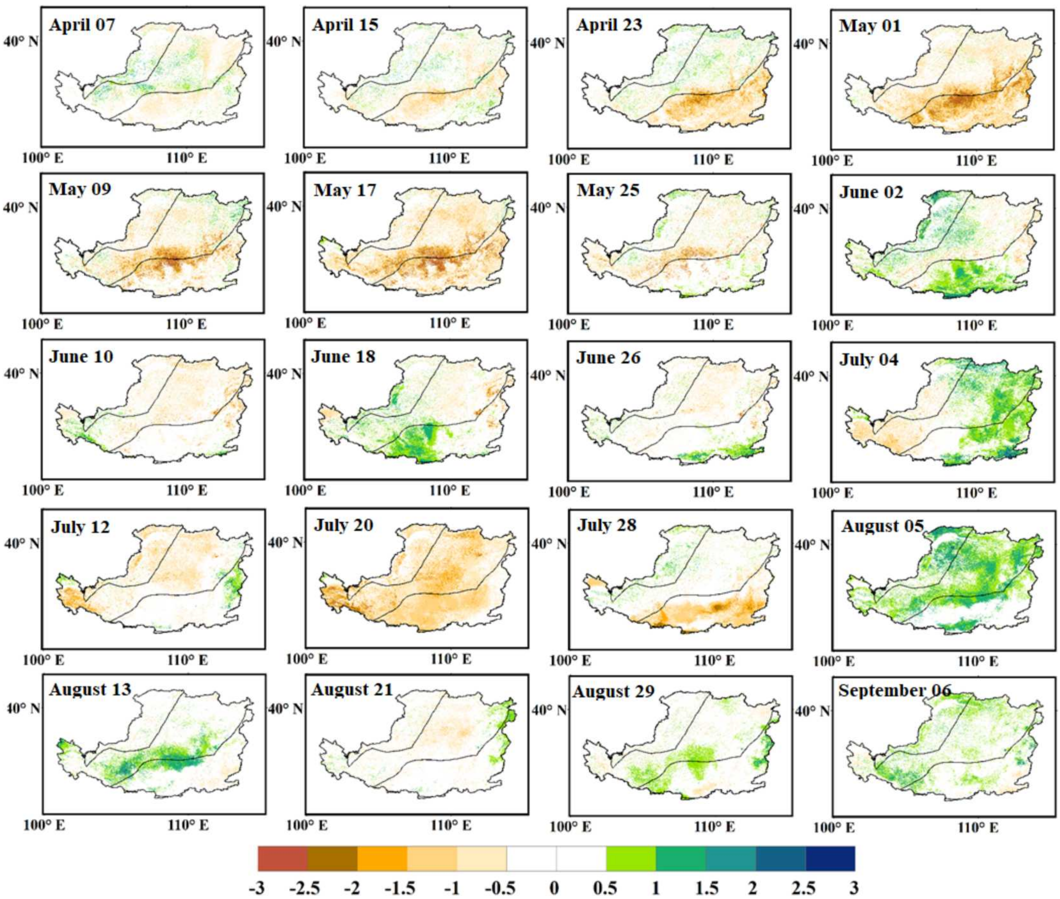

Figure 7. The 8-day GPP RCI values during the period from April 07 to September 06 in 2001.

detected negative values during drought development, it was less synchronized with the drought detected by SPEI, and it also did not match the drought region and time as detected by the SIF-RCI. Further, the synchronization of change curves of the minimum and average values between SIF-RCI and SPEI in the two sample areas was significantly better than that between GPP and SPEI (Figure 9). Compared with SIF-RCI, GPP RCI is less sensitive in indicating drought development, so GPP RCI cannot provide a reliable drought warning for the onset and progression of drought.

3.5. The response of GOSIF, GPP and hydrometeorological indicators to drought event

Spatial distribution changes of monthly GPP standardized anomalies resembled those of GOSIF in most of the LP. GOSIF and GPP were both abnormally high in most of the LP in March, especially in the zones of S-H and H, but the positive anomalies decreased from April to May and fell below normal by June in most of the LP (Figure 10). The PPT first decreased from April and gradually increased from July. The negative anomalies of PPT also led to negative anomalies in ET, coupled with vegetation growth in early spring, which undoubtedly accelerated the consumption of SM. Therefore, the SM in most of the LP was below normal from May to June. The lack of water led to poor vegetation growth from May to July, resulting in negative anomalies of GPP and GOSIF.

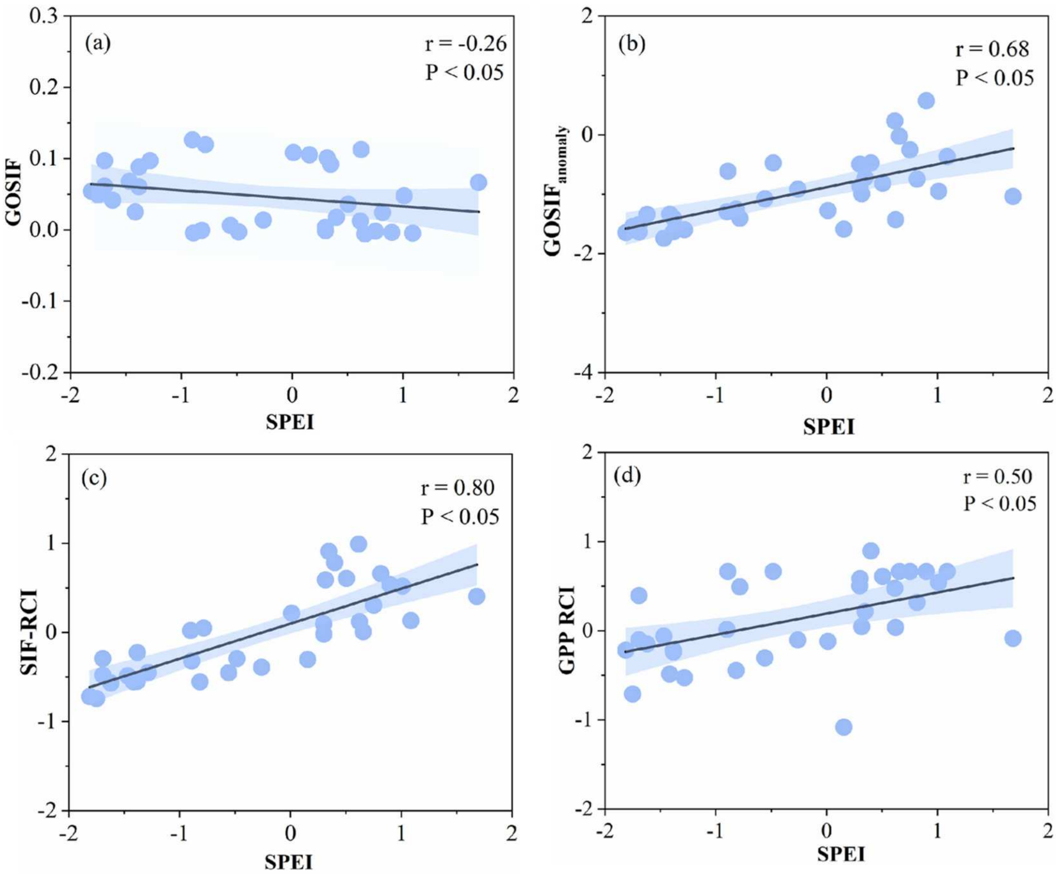

Figure 8. The correlation coefficients between GOSIF value, GOSIF normalized anomaly, SIF-RCI, GPP RCI and SPEI over the entire LP.

SM also increased with rising PPT from July. In August, with a significant increase in PPT, T also fell below normal. The improvement of meteorological conditions promoted the vegetation growth, so the GPP and GOSIF rose above normal by August. Correspondingly, as the intensity of drought weakened in July, the negative value of SIF-RCI also gradually disappeared, and the positive area of SIF-RCI gradually increased (Figure 6).

To determine the correlation between environmental factors and vegetation photosynthesis during the 2001 drought, we analyzed the correlation between GOSIF and hydrometeorological parameters of T, PPT and PAR. The results showed the correlation between GOSIF and T was the strongest during droughts ($r = 0.88$, $P < 0.05$), followed by PPT, and the association between PAR and GOSIF was not significant (Figure S4). Figure 11 shows the correlation coefficients between T, PPT, PAR, SM and GOSIF in different climate zones and vegetation types from 2001 to 2020. As the drought intensity increases, the impacts of PPT and SM on GOSIF gradually increase, but the impacts of T and PAR on GOSIF gradually weaken (Figure 11(a)). Furthermore, the impacts of PPT and SM on GOSIF gradually increase for forests, farmland, grasslands, and deserts, but T and PAR are the opposite (Figure 11(b)).

4 Discussion

4.1. Potential assessment of SIF-RCI for drought warning in the LP

In recent years, SIF has become a novel data source to monitor regional vegetation growth conditions under water stress (Verma et al. 2017; Liu, Dang, and Yue 2021). The SIF data type

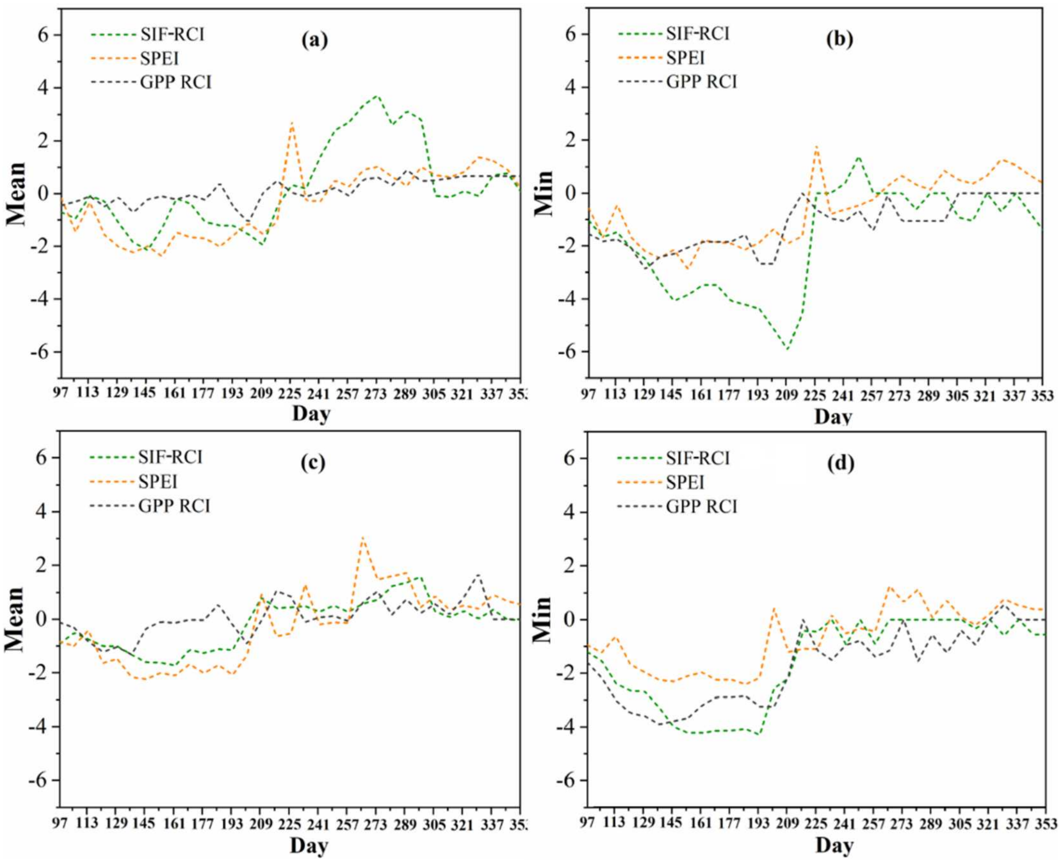

Figure 9. The 8-day time series of spatially averaged SIF-RCI, SPEI and GPP RCI in 2001 in the sample area 1 (a and b) and sample area 2 (c and d) of LP. The scale on the x-axis represents the day of the year.

commonly used in most studies was the SIF value or SIF anomaly (Li et al. 2022; Liu, Dang, and Yue 2021). In this study, in addition to the SIF value or SIF anomaly, we also used the RCI calculation formula introduced by Otkin et al. (2014) in GOSIF and MODIS-GPP to quantify the degree of unusual rates of change. By comparing the performance of satellite data directly related to vegetation photosynthesis during drought, we evaluated their drought warning potentials. Mohammadi, Jiang, and Wang (2022) has proven the hypothesis that the SIF trajectory served as an early warning for the onset of flash drought if it increased faster or decreased slower than usual in the early stage of drought. Flash drought is a subset of drought, referring to rapidly developing drought events that may eventually develop into slow droughts (Svoboda et al. 2002). Therefore, the potential of SIF-RCI to provide early warning for drought is beyond doubt.

To test the performances of GOSIF value, GOSIF anomaly, and SIF-RCI as drought warning, we took a recorded extreme drought event in the LP in 2001 as an example and selected the widely used meteorological drought index SPEI as a comparison reference (Zhang et al. 2018; 2015). Here, we found that GOSIF value described the growth status of vegetation, which gradually increased with vegetation growth during the growing season, making it difficult to quickly identify changes in vegetation growth caused by drought (Figure S2). The GOSIF anomaly remained below the normal level throughout the entire growth period (Figure S3), and the vegetation in A and S-A zones suffered the most severe drought before July, which was attributed to insufficient water in the LP. Water was the main factor limiting vegetation growth in LP; therefore vegetation was highly susceptible to water stress (Feng et al. 2016). Although the deviation between vegetation and normal

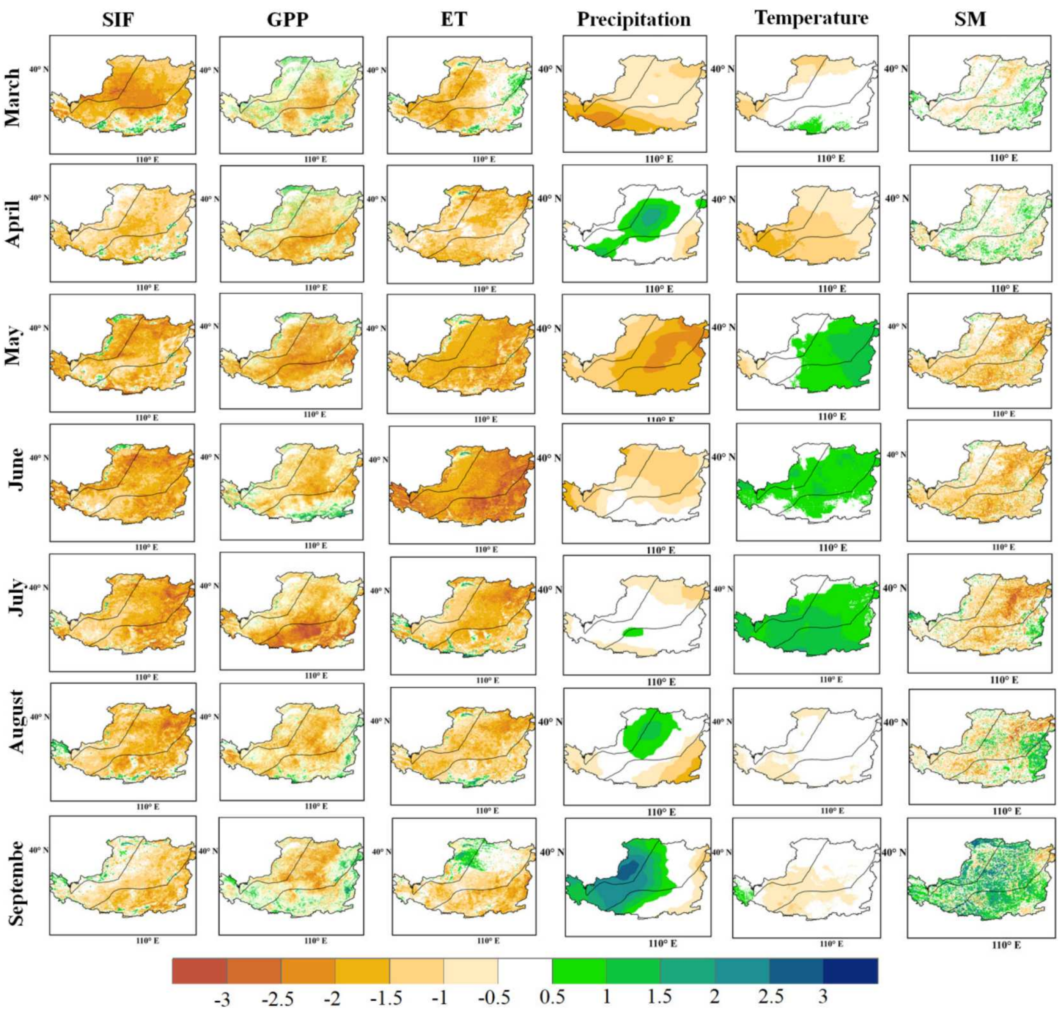

Figure 10. Standardized anomalies of monthly GOSIF, GPP, ET, T, PPT, and SM prior to and during the 2001 drought.

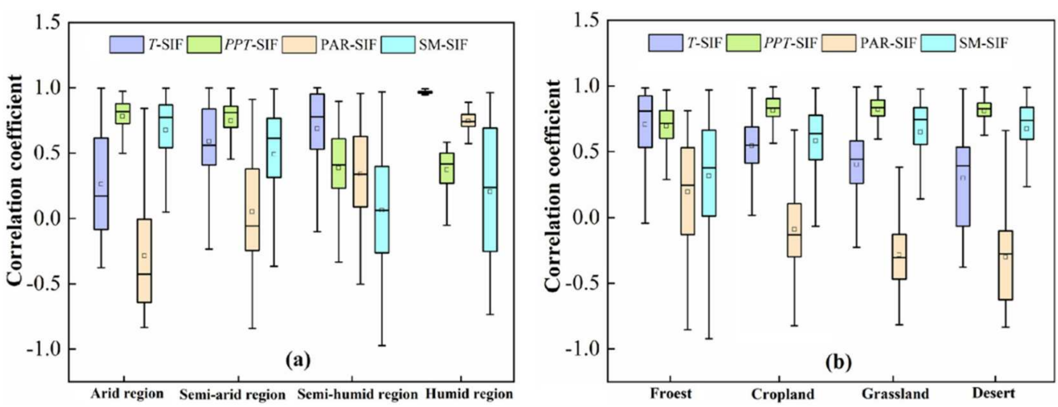

Figure 11. Variation of correlation coefficients between T, P, PAR, SM and GOSIF in different climatic zones (a) and different vegetation types (b) in the LP of China.

growth conditions during drought were captured by GOSIF anomaly, the dynamic information of the temporal change rate of vegetation photosynthesis had not been clearly identified.

By contrast, the large SIF-RCI negative values could not only detect drought onset almost simultaneously with the SPEI, but also disappeared synchronously with SPEI drought signals when the drought ended (Figure 6). Moreover, the drought intensity of SPEI and the negative SIF-RCI signals peaked simultaneously on 25 May, and then the negative SIF-RCI signals decreased with the weakening of drought intensity and disappeared completely when drought ended (Figures 3 and 6). But notably, unlike SPEI, SIF-RCI did not indicate drought intensity. The SIF-RCI signals were stronger in the early stage of drought occurrence, and they gradually decreased once the drought was in its peak due to the weakening of photosynthetic activity. Another important finding was that the positive SIF-RCI signals could be detected at least 4 8-days periods before the end of drought, which showed a faster increase or a slower decrease in GOSIF than normal, indicating that vegetation was gradually recovering from the drought. Furthermore, our results showed that the SIF-RCI responded more strongly to meteorological drought than the GOSIF value and GOSIF anomaly did, as the correlation between SIF-RCI and SPEI was more significant than the GOSIF-SPEI and $\text{GOSIF}_{\text{anomaly}}$-SPEI correlations (Figure 8). The key advantage of SIF-RCI is its ability to synchronize with the drought intensity monitored by SPEI, providing timely warnings and tracking both the onset and recovery phases of drought. Unlike the GOSIF value and GOSIF anomaly, which focus on absolute values or deviations, SIF-RCI identifies rapid transitions making it more responsive to short-term changes in vegetation photosynthesis during drought. Therefore, SIF-RCI has a greater potential as a drought warning indicator than GOSIF value and GOSIF anomaly.

The significant correlation between the maximum SIF-RCI signals and the maximum GOSIF values and the maximum SIF-RCI signals and SPEI in Figure 12 from 2001 to 2020 is sufficient to demonstrate the credibility of SIF-RCI as a drought warning. Although there were three false alarms in sample area 1 (occurring in 2016, 2017, and 2018) and one false alarm in sample area 2 (occurring in 2015) (Figure 12(a) and (c)), the SIF-RCI of other years exhibited a significant correlation with the SPEI signals. Furthermore, the predicted results of the red and green dashed lines in Figure 12 indicated that the correlation between SIF-RCI and SPEI was largely unaffected by the extreme drought event of 2001. Therefore, SIF-RCI has great potential as an early warning index of drought occurrence and recovery, which is of great significance for improving the accuracy of vegetation physiological drought monitoring.

In addition to the inconsistency of the coverage area and development trajectory with negative SIF-RCI signal, we also showed additional evidence that the negative GPP RCI values were not well synchronized with the drought signal detected by SPEI (Figure 7), thus the GPP RCI was not as effective as the SIF trajectory in reflecting drought development. The reason for this phenomenon may be related to the data source used. Instead of the original SIF data, the data adopted in this study was the GOSIF product reconstructed by a machine learning algorithm based on vegetation index and meteorological data, thus containing uncertain factors (Li et al. 2022). But notably, Mohammadi, Jiang, and Wang (2022) also obtained the same result by using the satellite-derived SIF observation data downscaled from GOME2. Therefore, the inconsistency between GPP RCI and SIF-RCI may be attributed to GPP data. In theory, above-ground GPP of vegetation is closely related to stomatal conductance and plant photosynthesis (Bai et al. 2022), but it was derived from reflectance data of MODIS and meteorological data rather than direct measurements. The mechanisms and reasons behind this relationship need to be further investigated but it is beyond the scope of this study.

Due to the close relationship between SIF and vegetation photosynthesis, we used GOSIF value and GOSIF-derived parameters in this study as indicators for drought warning. Vegetation itself does not have causal sources that can warn of drought, instead, it responds through photosynthesis to different environmental changes when it is subjected to water stress. Consequently, drought can be tracked from the changed trajectory of SIF during drought. In fact, no matter what kind of meteorological drought index is used to monitor drought, the impact scope and severity of drought

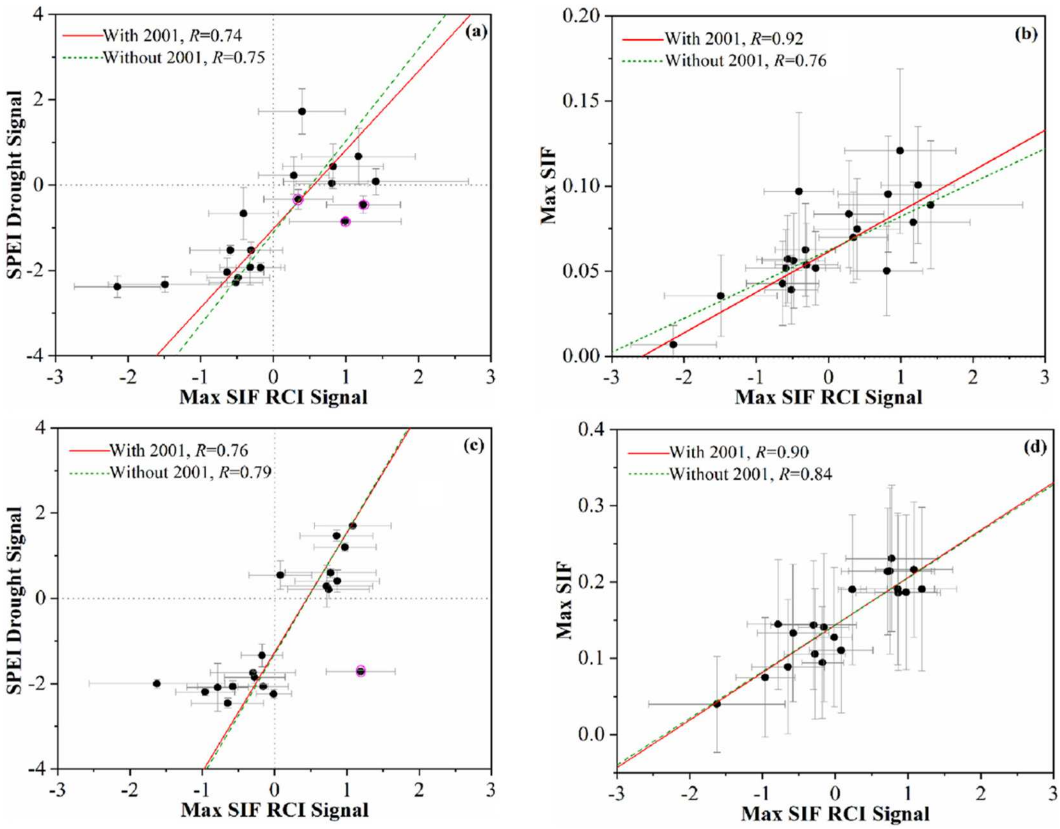

Figure 12. Scatter plot between maximum GOSIF (b and d), SPEI (a and c) and the maximum signal value of SIF-RCI in May–June of 2001-2020.

are only determined from environmental factors, but vegetation is the ultimate target of drought. Therefore, we should shift the focus from meteorological conditions to the quality of vegetation growth. Space-borne SIF data directly related to vegetation photosynthesis will be the main data source in future studies of global vegetation ecosystem response to global climate change, especially for remote areas lacking sufficient meteorological data.

4.2. Effects of hydrometeorological factors on GOSIF

Our results demonstrated the large negative SIF-RCI values observed from the end of April to May were due to the exposure of vegetation to multiple environmental stresses. The appearance of positive anomalies in GPP and GOSIF in March suggested that vegetation had begun to grow before the normal growing season (Figure 8). The growth of vegetation inevitably increased the consumption of water, thus speeding up soil moisture loss. Subsequent lack of sufficient precipitation may alter vegetation growth rates, thereby reducing GPP and GOSIF. However, the absence of precipitation was not the only factor leading to the rapid deterioration of vegetation growth. The abnormally high temperature and evapotranspiration along with low photosynthetically active radiation may also contribute to the abnormal vegetation growth. The significant correlation in Figure S3 between GOSIF and temperature and soil moisture during drought indicated that the increase in temperature and the decrease in soil moisture exacerbated the changes in GOSIF. Li and Xiao (2020) demonstrated that GOSIF was more sensitive to changes in soil moisture, and soil moisture availability had a strong control effect on the productivity of dryland ecosystems. The potential of SIF-RCI as a drought warning was increased by the stress response of vegetation photosynthesis under various stress environments.

The effects of meteorological factors on vegetation photosynthesis on the LP are different in different climate zones and vegetation types. The impacts of PPT and SM on GOSIF gradually rose as drought intensity increased, but the impacts of T and PAR gradually weakened (Figure 11(a)), which was consistent with Li and Xiao (2020). Hence, water regulated the photosynthetic activity of vegetation to a large extent, especially in the arid environment, which was an important factor affecting the photosynthesis of vegetation in low altitude areas. Furthermore, the impacts of PPT and SM on GOSIF gradually increased for forests, farmland, grasslands, and deserts, but T and PAR were the opposite (Figure 11(b)). Forest is the most resistant to drought stress, followed by cropland, but grassland and desert are the weakest, which is consistent with the research results of Xu et al. (2018). The vegetation in the zones of A and S-A is mainly composed of grassland and desert, with shallow root depths. The woody system of herbaceous plants has a low ability to store water and carbon, resulting in poor resistance to drought stress. Their physiological activity is determined by surface soil moisture and is therefore highly sensitive to rainfall dynamics (Kulmatiski and Beard 2013). Therefore, vegetation in A and S-A zones is the most sensitive to drought response, and water has the greatest impact on vegetation photosynthesis compared with other climate zones. Compared with grassland, forest roots are well developed and can store water in the deep soil, and in a dry environment, the roots are able to use the water stored in the deep soil in advance, thereby alleviating the effects of drought stress (McDowell et al. 2008). Unlike grasslands and woodlands, the photosynthetic response to drought in farmland can be modulated through agricultural management measures such as irrigation, fertilization, and rotation (Frankenberg et al. 2011). Especially, the response of vegetation in humid areas to water scarcity is not as sensitive as that of vegetation with the same biotic community type in relatively arid areas (Xu et al. 2020). This may be attributed to vegetation in the S-A zone slowing down photosynthesis during drought by closing stomata, thereby maintaining stem water potential (Xu et al. 2020; van der Molen et al. 2011).

4.3. Suggestions for future studies

The ability of SIF to accurately capture the response process of vegetation to water stress makes it a suitable physiological signal for early detection and monitoring of regional drought and provides valuable information and effective methods for disaster early warning (Ma et al. 2018). SIF-RCI derived from the satellite of GOME-2 has already demonstrated its early warning potential in two flash drought events in the United States (Mohammadi, Jiang, and Wang 2022). In this study, by comparing it with SPEI, we have demonstrated that SIF-RCI is essentially compatible with SPEI in detecting the extent, occurrence and evolution of drought events. Yet, SPEI is generally used as a traditional drought monitoring indicator and few studies have reported that SPEI has a flash drought warning effect. So, whether the flash drought function of SPEI also has applicability in the LP of China, or whether SPEI has the potential for indicating flash drought needs further investigation. Furthermore, rather than using the direct SIF observation data, the GOSIF data used in this study was developed based on the data with discrete coarse resolution (OCO-2 SIF, MODIS, MERRA-2). The coarse spatiotemporal resolution prevents SIF from being used for accurately detecting the photosynthetic activity of terrestrial vegetation. Therefore, there are certain inherent uncertainties in the data of GOSIF products, and the response of physiological processes such as vegetation photosynthesis to changes in water and heat conditions may vary with changes in vegetation phenology, resulting in deviations when analyzing the relationship between SIF and SPEI. Soon, more accurate all-day-long observations data with high spatiotemporal resolution for detecting photosynthetic activities of terrestrial vegetation are expected to be available. With the emerging new datasets, it is promising to further improve our understanding on vegetation photosynthesis and the applications in drought monitoring, ecosystem functions and terrestrial carbon cycle studies.

5. Conclusion

This study compared the performances of GOSIF value, GOSIF anomaly and SIF-RCI during typical drought events across China's LP and used SPEI as a drought reference to evaluate their potential to provide early warning of drought. The results showed that SIF-RCI had more reliable drought early warning potential in the LP compared with GOSIF value and GOSIF anomaly. The negative signal of SIF-RCI can be used to detect the occurrence of a drought event and it has good agreement with the results from the SPEI in the LP. The SIF-RCI positive signal appeared at least 4 8-day periods before the end of the drought indicating that vegetation had gradually begun to recover before the end of the drought. Furthermore, GPP RCI also showed negative signals during drought, but compared to SIF-RCI, GPP RCI signals were weaker and spatially dispersed so were unable to provide consistent predictors for drought onset. Meteorological factors had a critical impact on vegetation photosynthesis, and their correlation coefficients with GOSIF varied in different climate and vegetation types. The correlation between PPT and SM and GOSIF gradually increased with the rise of drought severity, and gradually increased from forest, farmland, grassland to desert, while the correlation between T and PAR and GOSIF was opposite. SIF-RCI can track the dynamics of vegetation changes caused by drought, demonstrating its advantages in accurately capturing the trajectory of drought development processes, such as onset and end, and is a promising alternative for regional drought monitoring and drought early warning.

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Disclosure statement

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Data availability statement

All data can be accessed via the corresponding author upon request.

CRediT authorship contribution statement

Yanmin Jiang: Conceptualization, Investigation, Methodology, Writing – original draft, Visualization. **Haijing Shi:** Supervision, Project administration, Methodology. **Xihua Yang:** Conceptualization, Writing – original draft, Funding acquisition. **Zhongming Wen:** Resources, Funding acquisition. **Youfu Wu:** Investigation, Writing – review and editing. **Ye Wang:** Investigation, Software. **Gaohui Duan:** Methodology, Writing – review and editing. **Angela G. Gormley:** Writing – review and editing. **Guang Yang:** Writing – review and editing. **Ji Li:** Writing – review and editing.

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